

PROJECT:

*Large Multimodal
Models for
Clinical Diagnosis*

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Module Info



Type of module

Project



Credit points

10



Weekly hours

4

Objectives



Work on practical research problems



Expand knowledge in *Healthcare, Medicine, Diagnostics, ...*



Deepen knowledge in *Computer Science, Artificial Intelligence, Large Language Models, ...*



Sharpen skills in *Software Engineering, Project Management, Teamwork, ...*



Strengthen interdisciplinary competencies in *Critical Thinking, Problem Solving, Decision Making, ...*

Content

<i>Large Multimodal Models for Tropical & Infectious Disease Diagnosis</i>	Tropical & Infectious Disease Diagnosis
	Large Language Models
	Multimodal Data
	Retrieval-Augmented Generation
	Finetuning

Deliverables



Diagnosis Support System for Tropical and Infectious Diseases using Multimodal Data

Clinical Diagnosis (Tropical and Infectious Cases)
Multimodal Data (Language + Image)
LMMs-Powered System (LLMs with Multimodality)



Project Management Documentation

Agile Software Development
DevOps



Technical Report

Usecases
Data
Models

Format

Teamwork

Class interaction

Weekly presentation + demo

Final presentation + demo

Final report

Grading



Continuous

30%



Final

70%

Hints

1. Data collection and processing
 - Text
 - Image
2. Methods in consideration
 - RAG
 - Finetuning
3. Model choices
 - CLIP/BLIP
 - Gemma/LLaVA
4. Optimization techniques
 - Quantization
 - PEFT, LoRA, QLoRA
5. Evaluation frameworks
 - RAG evaluation
 - Finetuning evaluation

References

Survey article

- Large Language Models for Disease Diagnosis: A Scoping Review

Research article

- Contextual Evaluation of Large Language Models for Classifying Tropical and Infectious Diseases

Data from NEJM

- Clinical Cases in Tropical Medicine

Models from Huggingface

- <https://huggingface.co>

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